

**Table 1: Simulation tools for blockchain-based decentralised wireless (DeWi), shared airwaves, RFID, antenna design**

| Tool                                                | Feature dimension: RF and shared spectrum                                           | Feature dimension: DeWi and blockchain                         | Feature dimension: System and physical layer                |
|-----------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------|
| <b>GNU Radio</b>                                    | Full SDR control; real-time spectrum monitoring                                     | Python block integration for Web3.py (Oracle)                  | Signal processing library (FFT, filters, modulators)        |
| <b>Web3.py (Python)</b>                             | Interacts with GNU Radio and SoapySDR data streams                                  | Primary bridge: sends RF data (RSSI, ID) to smart contracts    | Native support for Ethereum/EVM-compatible chains           |
| <b>Remix IDE</b>                                    | Simulation of timed access token NFTs for frequencies                               | Write/debug Solidity contracts for spectrum leasing            | Browser-based; no-setup required for rapid testing          |
| <b>4nec2 (antenna design)</b>                       | Simulation of MIMO beamforming and phase steering (coherent EX)                     | Model geometry for decentralised distributed MIMO arrays       | Far-field radiation patterns and impedance plots            |
| <b>NS-3 (network simulator)</b>                     | Advanced simulation of LTE, 5G NR, and Wi-Fi coexistence                            | Contributed modules for basic blockchain framework simulation  | Packet-level simulation for massive DeWi node counts        |
| <b>OpenEMS Finite-Difference Time-Domain (FDTD)</b> | High performance toolkit; 3D full-wave electromagnetic solver (ECG Cartesian grids) | Simulates RFID antenna-tag interaction in complex environments | Matlab/Octave interface for scripting and processing        |
| <b>Helium Console/Explorer</b>                      | View actual LoRaWAN and 5G hot spot coverage data                                   | Real-time visualisation of proof-of-coverage (PoC) events      | Monitor data packet transfers and token rewards             |
| <b>Tatum (API)</b>                                  | 'Blockchain-as-a-Service'; easy to link SDR apps                                    | Generous free tier for simplified blockchain API calls         | Abstracted SDKs for 40+ different protocols (BTC, ETH, LTC) |
| <b>Truffle/Hardhat</b>                              | Local testnets (Ganache) to simulate instant spectrum transactions                  | Frameworks for deploying complex multi-contract DeWi protocols | Automated testing pipeline for smart contract security      |
| <b>CST Studio Suite</b>                             | Simulate human body effects on wearable DeWi devices                                | Model CBRS/5G antennas with precise physical constraints       | Coupled EM-thermal solver for high-power transmitters       |
| <b>Dune Analytics</b>                               | Monitor active node counts and data usage (e.g., Helium dashboards)                 | SQL-based tool to visualise DeWi network token economics       | Track DeWi market trends and growth                         |
| <b>MATLAB (RF Toolbox)</b>                          | Simulate full RF chains (budget analysis, mixers, power amps)                       | Link RF simulations to distributed ledger prototypes           | Support for standards (5G, LTE, WLAN, satellite)            |
| <b>Wireshark</b>                                    | Decode raw RF protocol packets (Zigbee, Bluetooth, custom LoRa)                     | Analyze Web3.py JSON-RPC calls between SDR and chain           | Troubleshoot data latency in decentralised networks         |
| <b>OpenRAN Gym</b>                                  | Simulate RAN Intelligent Controllers (RIC) with SDR loops                           | Evaluate decentralised O-RAN architecture components           | Toolbox for applying AI/ML to spectrum sharing              |